

IF440: Artificial Intelligence

Topic 03: First Order Logic

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Course Sub Learning Outcomes (Sub-CLO):



SubCLO-0221: Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing (C3).

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Review

- 1 Concept of Knowledge representation
- 2 How can knowledge be represented?
- 3 Propositional logic
- 4 Horn Clauses



Outline

- 1 First-Order Logic
- 2 Syntax of FOL
- 3 Quantifiers
- 4 Assertions and Queries
- 5 Inference in FOL



Pre-Class Assignment



Watch this video tutorial:

- Stanford Online https://www.youtube.com/watch?v=Z-00Q3_oTJM

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First-Order Logic

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FIRST ORDER LOGIC (FOL)

- AKA **Predicate Logic**.
- First-order logic is a logical system for reasoning about properties of objects.
- Predicate logic is an extension of propositional logic.
- Augments the logical connectives from propositional logic with
 - **predicates** that describe properties of objects,
 - **functions** that map objects to one another, and
 - **quantifiers** that allow us to reason about multiple objects.

FOL vs PL

- PL does not signify or express the generalization, specialization or pattern for example '**QUANTIFIERS**' cannot be used in PL but in FOL users can easily use quantifiers as it does express the generalization, specialization, and pattern.
- Propositional Logic converts a complete sentence into a **symbol** and makes it logical whereas in First-Order Logic relation of a particular sentence will be made that involves **relations, functions, and constants**.

FIRST ORDER LOGIC (FOL)

- First-Order Logic assumes that the world contains:
 - **Constants**
E.g. IF607, B307, John, Teacher, apple, ...
 - **Relations/predicates**
E.g. is_Man(John), Located(IF607, B307), is_kind_of(apple, fruit),
brother_of(John, Richard)
 - **Functions** – relations in which there is only one value for a given input.
E.g. father_of(), best_friend_of(), ...

Predicates

- To reason about objects, first-order logic uses **predicates**.
- Examples:
 - Cute(Koala)
 - Likes(children, candies)
- Applying a predicate to arguments produces **a proposition**, which is either **true** or **false**.
- Typically, when you're working in FOL, you'll have a list of predicates, what they stand for, and how many arguments they take. It'll be given separately than the formulas you write.



Equality

- First-order logic is equipped with a special predicate $=$ that says whether two objects are equal to one another.
- Equality is a part of first-order logic, just as $\rightarrow$ and $\neg$ are.
- Examples:
 - $\text{TomMarvoloRiddle} = \text{LordVoldemort}$
 - $\text{MorningStar} = \text{EveningStar}$
- **Equality** can only be applied to **objects**; to state that two propositions are equal, use $\leftrightarrow$.

Functions

- First-order logic allows **functions** that return **objects** associated with **other objects**.
- Examples:
 - ColorOf(Money)
 - MedianOf(x, y, z)
 - $x + y$
- As with predicates, functions can take in any number of arguments, but always return **a single value**.
- Functions evaluate to **objects, not propositions**.

The Type-Checking Table

	... operate on ...	... and produce
Connectives ($\leftrightarrow$, $\wedge$, etc.) ...	propositions	a proposition
Predicates ($=$, etc.) ...	objects	a proposition
Functions ...	objects	an object

$Sentence \rightarrow AtomicSentence \mid ComplexSentence$
 $AtomicSentence \rightarrow Predicate \mid Predicate(Term, \dots) \mid Term = Term$
 $ComplexSentence \rightarrow (Sentence) \mid [Sentence]$
 $\quad \mid \neg Sentence$
 $\quad \mid Sentence \wedge Sentence$
 $\quad \mid Sentence \vee Sentence$
 $\quad \mid Sentence \Rightarrow Sentence$
 $\quad \mid Sentence \Leftrightarrow Sentence$
 $\quad \mid Quantifier Variable, \dots Sentence$

$Term \rightarrow Function(Term, \dots)$
 $\quad \mid Constant$
 $\quad \mid Variable$
 $Quantifier \rightarrow \forall \mid \exists$
 $Constant \rightarrow A \mid X_1 \mid John \mid \dots$
 $Variable \rightarrow a \mid x \mid s \mid \dots$
 $Predicate \rightarrow True \mid False \mid After \mid Loves \mid Raining \mid \dots$
 $Function \rightarrow Mother \mid LeftLeg \mid \dots$

OPERATOR PRECEDENCE : $\neg, =, \wedge, \vee, \Rightarrow, \Leftrightarrow$

Syntax of FOL



SYNTAX OF FOL

- **Constant Symbols:** *KingJohn, 2, ...*
- **Predicate symbols:** *Brother, >, ...*
- **Function Symbols:** *Sqrt, LeftLegOf, ...*
- **Variable symbols:** *x, y, a, b, ...*
- **Connectives:** $\wedge, \vee, \neg, \Rightarrow$
- **Equality:** $=$
- **Quantifiers:** $\forall, \exists$
- **Punctuation:** $() ,$

SYNTAX OF FOL

- **Term:** $function(term_1, ..., term_n) \mid constant \mid variable$
 $LeftLeg(John)$
- **Atomic sentences:** $predicate(term_1, ..., term_n) \mid term_1 = term_2$
 $Brother(Richard, John); Married(Father(Richard), Mother(John))$
- **Complex sentences:** $Brother(John, Richard) \wedge Brother(Richard, John)$ or
 $\neg King(Richard) \Rightarrow King(John)$

SYNTAX OF FOL

- **Universal Quantifiers:** $\forall x \text{ King}(x) \Rightarrow \text{Person}(x)$
 - The expression is true for **every possible value** of the variable.
 - “For all...”, “All...”
- **Existential Quantifiers:** $\exists x \text{ Crown}(x) \wedge \text{OnHead}(x, \text{John})$
 - The expression is true for **at least one value** of the variable.
 - “There exists...”, “There is at least one...”, “For some...”
- **Equality:** $\text{Father}(\text{John}) = \text{Henry}; \exists x, y \text{ Brother}(x, \text{Richard}) \wedge \text{Brother}(y, \text{Richard}) \wedge \neg(x = y)$
 - To **state facts** about a given function.
 - To signify that two terms **refer to the same object / not.**

IMPLICATION TRUTH TABLE

α	β	$\alpha \Rightarrow \beta$
T	T	T
T	F	F
F	T	T
F	F	T

- If α is true, then β must be true for the implication to be true
- If α is false, then we have no information about β
 - β might be true or false without affecting the implication
 - β is *not* necessarily true
- If implication is true and α is true, then β must be true (Modus Ponens)

Quantifiers



UNIVERSAL IMPLICATION

$x = \{ \text{🍄}, \text{🍄}, \text{🤖} \}$

All purple mushrooms are poisonous

- $\forall x \text{ Purple}(x) \wedge \text{Mushroom}(x) \Rightarrow \text{Poisonous}(x)$

Universal: all three implications are true

- 🍄 is purple and a mushroom $\Rightarrow$ is poisonous
- 🍄 is purple and a mushroom $\Rightarrow$ is poisonous
- 🤖 is purple and a mushroom $\Rightarrow$ is poisonous

But the first one is the only one where the premise is true, therefore the only one where we know the consequent is true (Modus Ponens)

- Otherwise, we don't know whether x is poisonous

EXISTENTIAL IMPLICATION

$x = \{ \text{🍄}, \text{🍄}, \text{👾} \}$

Some purple mushrooms are poisonous

- $\exists x \text{ Purple}(x) \wedge \text{Mushroom}(x) \Rightarrow \text{Poisonous}(x)$

Existential: at least one implication is true

- 🍄 is purple and a mushroom $\Rightarrow$ is poisonous
- 🍄 is purple and a mushroom $\Rightarrow$ is poisonous
- 👾 is purple and a mushroom $\Rightarrow$ is poisonous

But all those where the premise is false are true

- The existential implication doesn't give us any information

EXISTENTIAL CONJUNCTION

$$x = \{ \text{🍄}, \text{🍄}, \text{🤖} \}$$

Some purple mushrooms are poisonous

- $\exists x \text{ Purple}(x) \wedge \text{Mushroom}(x) \wedge \text{Poisonous}(x)$

Existential: at least one statement is true

- 🍄 is purple and a mushroom and is poisonous
- 🍄 is purple and a mushroom and is poisonous
- 🤖 is purple and a mushroom and is poisonous

The first statement is true, the other two are false

- The existential quantifier is satisfied
- The situation is consistent with our original meaning

UNIVERSAL CONJUNCTION

$$x = \{ \text{🍄}, \text{🍄}, \text{👤} \}$$

All purple mushrooms are poisonous

- $\forall x \text{ Purple}(x) \wedge \text{Mushroom}(x) \wedge \text{Poisonous}(x)$

Universal: all three statements are true

- 🍄 is purple and a mushroom and is poisonous
- 🍄 is purple and a mushroom and is poisonous
- 👤 is purple and a mushroom and is poisonous

"Everything is a poisonous purple mushroom"

- This is false, and not the original meaning of our sentence

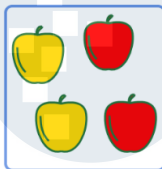
Universal and Existential Quantifier



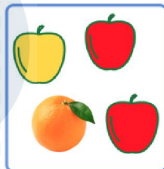
Some apples
are green



All apples
are green



No apples
are green



TRY THIS

- What is the interpretation for:
 1. $\text{King}(\text{Richard}) \vee \text{King}(\text{John})$
 2. $\text{In}(\text{Paris, France}) \wedge \text{In}(\text{Marseilles, France})$
 3. $\text{Mother}(\text{John}) = \text{Mother}(\text{Richard})$



TRY THIS

1. Dumbo is grey
 $\text{grey}(\text{dumbo})$
2. Elephants are grey
 $\forall x \text{ elephant}(x) \Rightarrow \text{grey}(x)$
3. There is a grey elephants
 $\exists x \text{ elephant}(x) \wedge \text{grey}(x)$





Assertions and Queries

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NESTING QUANTIFIERS

- Everyone likes some kind of food
 $\forall y \exists x, \text{Food}(x) \wedge \text{Likes}(y, x)$
- There is a kind of food that everyone likes
 $\exists x \forall y, \text{Food}(x) \wedge \text{Likes}(y, x)$
- Someone likes all kinds of food
 $\exists y \forall x, \text{Food}(x) \wedge \text{Likes}(y, x)$
- Every food has someone who likes it
 $\forall x \exists y, \text{Food}(x) \wedge \text{Likes}(y, x)$

PROPERTIES OF QUANTIFIERS

- $\forall x \forall y$ is the same as $\forall y \forall x$
- $\exists x \exists y$ is the same as $\exists y \exists x$
- $\exists x \forall y$ is not the same as $\forall y \exists x$
 - $\exists x \forall y \text{ Loves}(x, y)$
“There is a person who loves everyone in the world.”
 - $\forall y \exists x \text{ Loves}(x, y)$
“Everyone in the world is loved by at least one person.”

PROPERTIES OF QUANTIFIERS

Quantifier Duality:

- Not everyone like McDonalds
 $\neg(\forall x, \text{Likes}(x, \text{McDonalds}))$
 $\exists x \neg \text{Likes}(x, \text{McDonalds})$

$$\begin{aligned}\forall x \neg P &\equiv \neg \exists x P \\ \neg \forall x P &\equiv \exists x \neg P \\ \forall x P &\equiv \neg \exists x \neg P \\ \exists x P &\equiv \neg \forall x \neg P\end{aligned}$$

- No one likes McDonalds
 $\neg(\exists x, \text{Likes}(x, \text{McDonalds}))$
 $\forall x \neg \text{Likes}(x, \text{McDonalds})$

$$\begin{aligned}\neg(P \vee Q) &\equiv \neg P \wedge \neg Q \\ \neg(P \wedge Q) &\equiv \neg P \vee \neg Q \\ P \wedge Q &\equiv \neg(\neg P \vee \neg Q) \\ P \vee Q &\equiv \neg(\neg P \wedge \neg Q)\end{aligned}$$

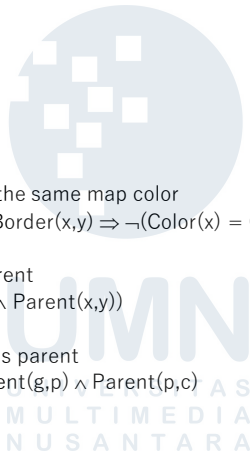
PROPERTIES OF QUANTIFIERS

- Universal quantifiers are often used with “implies” to form “rules”:
 $\forall x \text{ student}(x) \Rightarrow \text{smart}(x)$ means “all students are smart”
- Universal quantification is rarely used to make blanket statements about every individual in the world:
 $\forall x \text{ student}(x) \wedge \text{smart}(x)$ means “everyone is a student and is smart”
- Existential quantifiers are usually used with “and” to specify a list of properties about an individual:
 $\exists x \text{ student}(x) \wedge \text{smart}(x)$ means “there is a student who is smart”

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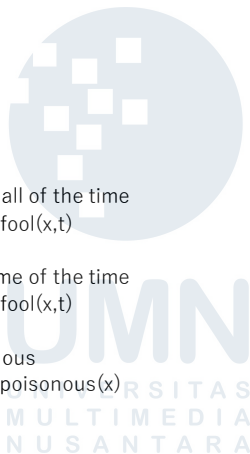
TRY THIS

- Brothers are siblings
 $\forall x,y \text{ Brother}(x,y) \Rightarrow \text{Sibling}(x,y)$
- No two adjacent countries have the same map color
 $\forall x,y \text{ Country}(x) \wedge \text{Country}(y) \wedge \text{Border}(x,y) \Rightarrow \neg(\text{Color}(x) = \text{Color}(y)) \wedge \neg(x = y)$
- One's mother is one's female parent
 $\forall x,y \text{ Mother}(x,y) \Leftrightarrow (\text{Female}(x) \wedge \text{Parent}(x,y))$
- A grandparent is a parent of one's parent
 $\forall g,c \text{ Grandparent}(g,c) \Leftrightarrow \exists p \text{ Parent}(g,p) \wedge \text{Parent}(p,c)$



TRY THIS

- Every gardener likes the sun
 $\forall x \text{ gardener}(x) \Rightarrow \text{likes}(x, \text{sun})$
- You can fool some of the people all of the time
 $\exists x \forall t \text{ person}(x) \wedge \text{time}(t) \Rightarrow \text{can-fool}(x,t)$
- You can fool all of the people some of the time
 $\forall x \exists t \text{ person}(x) \wedge \text{time}(t) \Rightarrow \text{can-fool}(x,t)$
- All purple mushrooms are poisonous
 $\forall x \text{ mushrooms}(x) \wedge \text{purple}(x) \Rightarrow \text{poisonous}(x)$
- Clinton is not tall
 $\neg \text{tall}(\text{Clinton})$





Inference in FOL

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INFERENCE IN FOL

■ **Universal Instantiation/Elimination**

KB:

- $\forall x \text{ King}(x) \wedge \text{Greedy}(x) \Rightarrow \text{Evil}(x)$
- $\text{King}(\text{John})$
- $\text{Greedy}(\text{John})$
- $\text{Brother}(\text{Richard}, \text{John})$

UI using $\{x/\text{John}\}$ and $\{x/\text{Richard}\}$

- $\text{King}(\text{John}) \wedge \text{Greedy}(\text{John}) \Rightarrow \text{Evil}(\text{John})$
- $\text{King}(\text{Richard}) \wedge \text{Greedy}(\text{Richard}) \Rightarrow \text{Evil}(\text{Richard})$

discard the Universally quantified sentence. We can get the KB to be propositions.

INFERENCE IN FOL

- **Existential Instantiation/Elimination**

- $\exists x \text{ Crown}(x) \wedge \text{OnHead}(x, \text{John})$
- $\text{Crown}(C1) \wedge \text{OnHead}(C1, \text{John})$

As long as C_1 does not appear elsewhere in the knowledge base. Must not belong to another object.

EXAMPLES OF INFERENCE RULES

- “It is illegal for students to copy music.”
- “Joe is a student.”
- “Every student copies music.”
- Is Joe a criminal?

Knowledge Base:

- $\forall x, y \text{ Student}(x) \wedge \text{Music}(y) \wedge \text{Copies}(x, y) \Rightarrow \text{Criminal}(x)$
- $\text{Student}(\text{Joe})$
- $\forall x \exists y \text{ Student}(x) \wedge \text{Music}(y) \wedge \text{Copies}(x, y)$

EXAMPLES OF INFERENCE RULES

From: $\forall x \exists y \text{ Student}(x) \wedge \text{Music}(y) \wedge \text{Copies}(x, y)$

↓ **Universal Elimination**

$\exists y \text{ Student}(\text{Joe}) \wedge \text{Music}(y) \wedge \text{Copies}(\text{Joe}, y)$

↓ **Existential Elimination**

$\text{Student}(\text{Joe}) \wedge \text{Music}(\text{SomeSong}) \wedge \text{Copies}(\text{Joe}, \text{SomeSong})$

↓ **Modus Ponens**

$\text{Criminal}(\text{Joe})$

BUILD AN INFERENCE ENGINE

- A very common behavior is to do:

1. And-Introduction

P, Q

$P \wedge Q$

2. Universal Elimination

3. Generalized Modus Ponens



BUILD AN INFERENCE ENGINE

Bob is a buffalo

Pat is a pig

Buffaloes outrun pigs

Bob outruns Pat

AI 1 & 2

UE 3, $\{x/Bob, y/Pat\}$

MP 4 & 5

1. $Buffalo(Bob)$

2. $Pig(Pat)$

3. $\forall x, y \text{ } Buffalo(x) \wedge Pig(y) \Rightarrow Faster(x, y)$

4. $Buffalo(Bob) \wedge Pig(Pat)$

5. $Buffalo(Bob) \wedge Pig(Pat) \Rightarrow Faster(Bob, Pat)$

6. $Faster(Bob, Pat)$

GENERALIZED MODUS PONENS

$$\frac{p_1', p_2', \dots, p_n', (p_1 \wedge p_2 \wedge \dots \wedge p_n \Rightarrow q)}{q\sigma} \quad \text{where } p_i'\sigma = p_i\sigma \text{ for all } i$$

E.g. $p_1' = \text{Faster}(\text{Bob}, \text{Pat})$

$p_2' = \text{Faster}(\text{Pat}, \text{Steve})$

$p_1 \wedge p_2 \Rightarrow q = \text{Faster}(x, y) \wedge \text{Faster}(y, z) \Rightarrow \text{Faster}(x, z)$

$\sigma = \{x/\text{Bob}, y/\text{Pat}, z/\text{Steve}\}$

$q\sigma = \text{Faster}(\text{Bob}, \text{Steve})$

- Two mechanisms for applying binding to Generalized Modus Ponens:
 - Forward chaining
 - Backward chaining
- In form of **definite clause**.

GENERALIZED MODUS PONENS

- The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold by Colonel West, who is American.
- Prove that Colonel West is a criminal!



KNOWLEDGE BASE

- ... it is a crime for an American to sell weapons to hostile nations:
1. $\text{American}(x) \wedge \text{Weapon}(y) \wedge \text{Sells}(x, y, z) \wedge \text{Hostile}(z) \Rightarrow \text{Criminal}(x)$
- Nono ... has some missiles i.e. $\exists x \text{Owns}(\text{Nono}, x) \wedge \text{Missile}(x)$:
2. $\text{Owns}(\text{Nono}, M1)$ and $\text{Missile}(M1)$
- ... all of its missiles were sold to it by Colonel West:
3. $\text{Missile}(x) \wedge \text{Owns}(\text{Nono}, x) \Rightarrow \text{Sells}(\text{West}, x, \text{Nono})$
- Missiles are weapons:
4. $\text{Missile}(x) \Rightarrow \text{Weapon}(x)$

KNOWLEDGE BASE

- An enemy of America counts as a “hostile”:
5. $\text{Enemy}(x, \text{America}) \Rightarrow \text{Hostile}(x)$
- West, who is American ...
6. $\text{American}(\text{West})$
- The country Nono, an enemy of America ...
7. $\text{Enemy}(\text{Nono}, \text{America})$

FORWARD CHAINING



American(West)

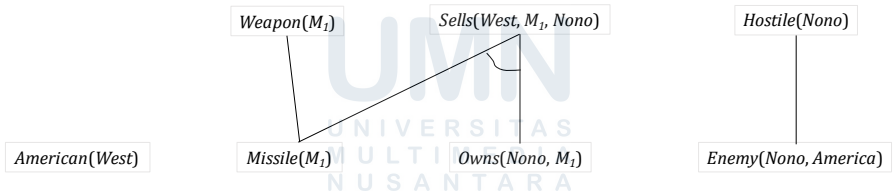
Missile(M_1)

Owns(Nono, M_1)

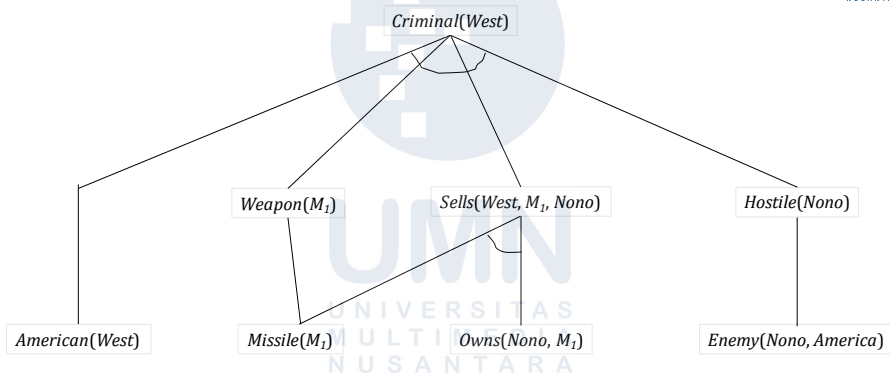
Hostile(Nono)

Enemy(Nono, America)

FORWARD CHAINING



FORWARD CHAINING



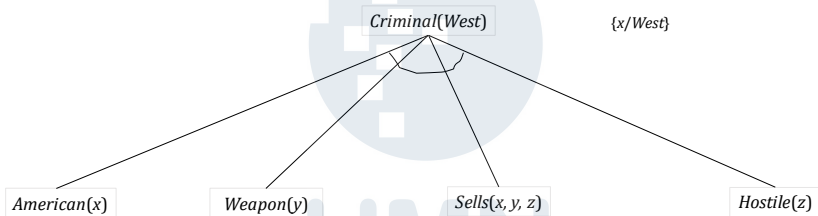
BACKWARD CHAINING



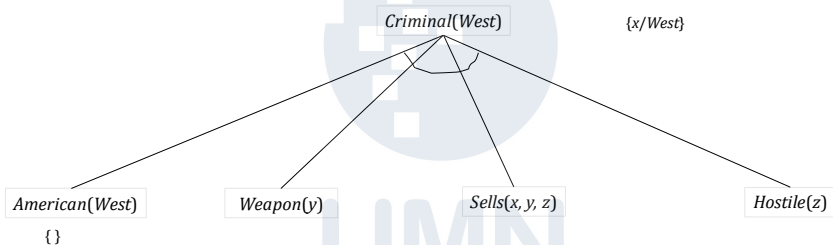
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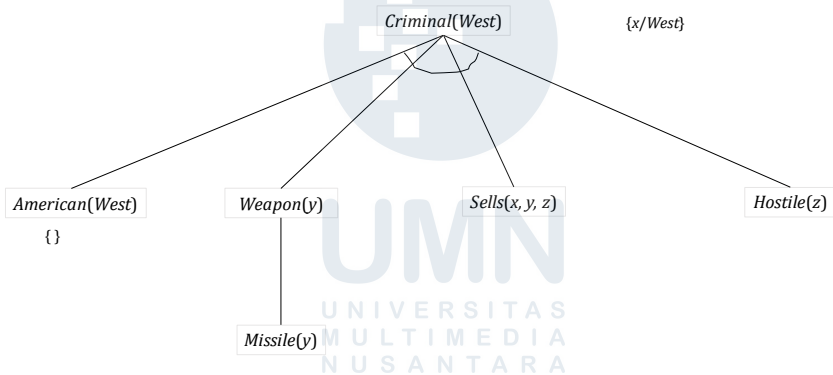
BACKWARD CHAINING



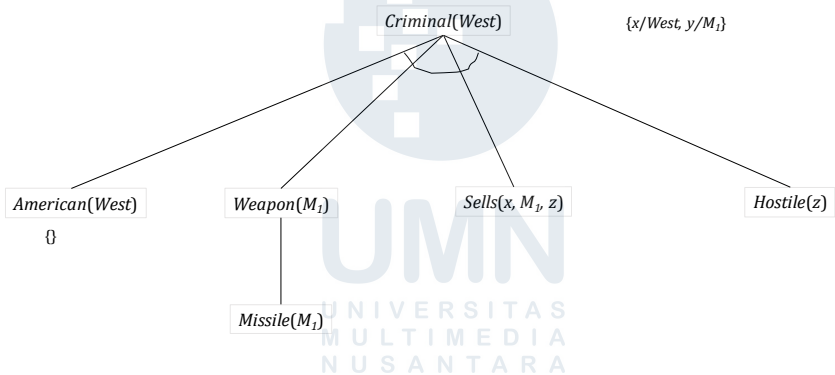
BACKWARD CHAINING



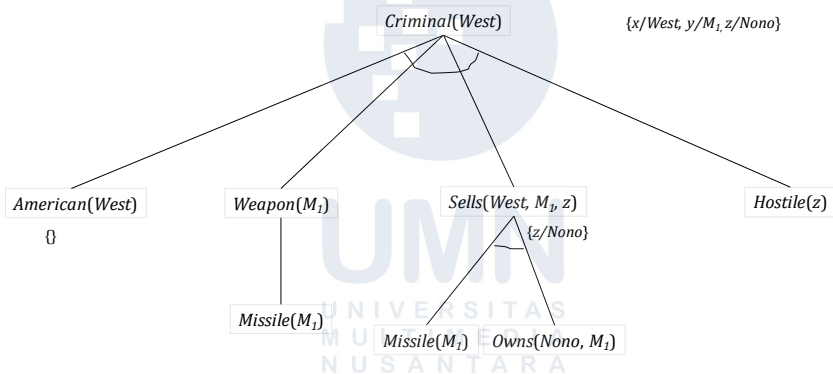
BACKWARD CHAINING



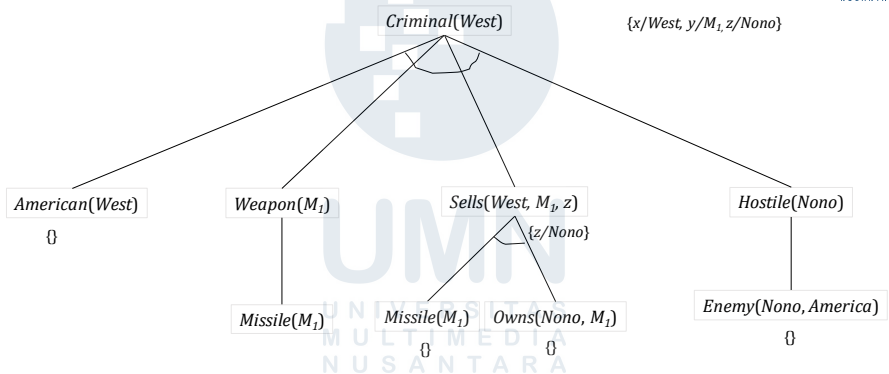
BACKWARD CHAINING



BACKWARD CHAINING



BACKWARD CHAINING



RESOLUTION

$$\frac{\ell_1 \vee \dots \vee \ell_k, \quad m_1 \vee \dots \vee m_n}{\ell_1 \vee \dots \vee \ell_{i-1} \vee \ell_{i+1} \vee \dots \vee \ell_k \vee m_1 \vee \dots \vee m_{j-1} \vee m_{j+1} \vee \dots \vee m_n}$$

■ Ex.:

$\neg \text{Rich}(x) \vee \text{Unhappy}(x)$

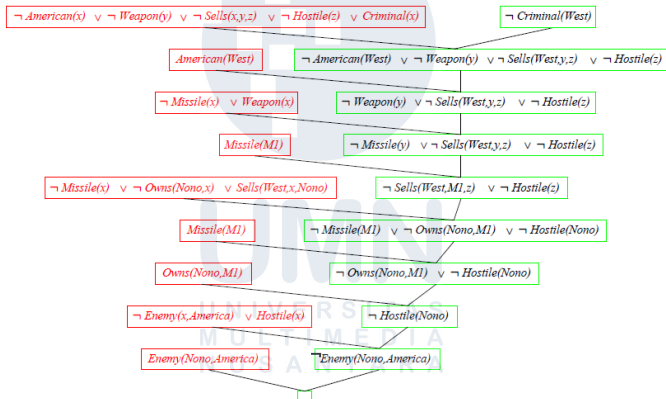
$\text{Rich}(\text{Ken})$

$\text{Unhappy}(\text{Ken})$

with $\theta = \{x/\text{Ken}\}$

*change to CNF (definite clause)

RESOLUTION



Exercise 1: Convert Sentences to FOL

■ Write these sentences in FOL:

- 1 All humans are mortal.
- 2 All cats love milk.
- 3 There exists at least one student who likes math.
- 4 Some dogs bark.
- 5 Someone walks and talks.



Exercise 2: Deduction in FOL

■ Premises:

- 1 All humans are mortal:

$$\forall x (Human(x) \rightarrow Mortal(x))$$

- 2 Socrates is a human:

$$Human(Socrates)$$

■ Goal: We want to deduce that Socrates is mortal:

$$Mortal(Socrates)$$

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Exercise 3: Deduction in FOL Using Negation and Proof by Contradiction

■ Premises:

- 1 All students who do not submit their assignments on time will fail the course:

$$\forall x(\text{Student}(x) \wedge \neg \text{SubmitsOnTime}(x) \rightarrow \text{Fails}(x))$$

- 2 ALICE is a student who passed the course:

$$\text{Student}(\text{ALICE}) \wedge \neg \text{Fails}(\text{ALICE})$$

- ### ■ Goal: We want to deduce that ALICE submitted her assignments on time:

$$\text{SubmitsOnTime}(\text{ALICE})$$

References

- [1] S. J. Russell, P. Norvig, and E. Davis, **Artificial intelligence: a modern approach**, 3rd ed. Prentice Hall, 2010.



Next Week

- Using uncertain knowledge
- Probability
- Conditional Probability
- Joint Probability Distribution
- Bayes' Theorem



Vision and Mission

Vision

Becoming a leading Informatics Undergraduate Study Program that produces graduates with international insight who are competent in the field of computer science, have an entrepreneurial spirit, and have noble character.



Mission

- 1 Organising learning with the best technology and curriculum and supported by professional teaching staff.
- 2 Carry out research activities in the field of informatics to advance information science and technology.
- 3 Carry out community service activities based on information technology and science in the context of practising informatics science and technology society.

